King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam II Term 222

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int \frac{x^2}{\sqrt{3-x^2}}\,dx =$

(a) $\frac{3}{2}\sin^{-1}\left(\frac{x}{3}\right) + \frac{\sqrt{3-x^2}}{x} + C$

(b) $\frac{3}{2}\sin^{-1}\left(\frac{x}{\sqrt{3}}\right) - \frac{1}{2}x\sqrt{3-x^2} + C$

(c) $\frac{\sqrt{3}}{2}\sin^{-1}\left(\frac{x}{3}\right) - \frac{1}{2}\sqrt{3-x^2} + C$

(d) $\frac{1}{\sqrt{3}}\sin^{-1}\left(\frac{x}{\sqrt{3}}\right) - \frac{3}{2}x\sqrt{3-x^2} + C$

(e) $\frac{1}{2}\sin^{-1}\left(\frac{x}{\sqrt{3}}\right) - x\sqrt{3-x^2} + C$

2. $\int \frac{\cos^3 t}{\sqrt{\sin t}}\,dt =$

(a) $\frac{\cos t}{\sqrt{\sin t}} - \frac{1}{5}\sin^3 t + C$

(b) $2\sqrt{\sin t} - \frac{2}{3}\sqrt{\sin^3 t} + C$

(c) $\frac{2\cos t}{\sqrt{\sin t}} - \frac{2}{5}\sqrt{\sin^3 t} + C$

(d) $2\sqrt{\sin t} - \frac{2}{5}\sqrt{\sin^5 t} + C$

(e) $\sqrt{\sin t} + 2\sqrt{\sin^5 t} + C$

3. Using the shell method, the volume of the solid obtained by revolving the region bounded by the curves $x = 4 - (y+1)^2$, $x = 0$ about the line $y = 2$ is given by

(a) $2\pi \int_{-3}^{1} (1+y)(2+y)(3+y)\,dy$

(b) $2\pi \int_{-3}^{1} (1+y)(2-y)(3+y)\,dy$

(c) $2\pi \int_{-3}^{1} (1-y)(2+y)(3-y)\,dy$

(d) $2\pi \int_{-3}^{1} (1-y)(2-y)(3+y)\,dy$

(e) $2\pi \int_{-3}^{1} (1+y)(2-y)(3-y)\,dy$

4. $\int_0^1 \frac{1}{(x^2+1)^{3/2}}\,dx =$

(a) $\dfrac{1}{\sqrt{2}}$

(b) $\sqrt{2}$

(c) $1 - \dfrac{1}{\sqrt{2}}$

(d) $\sqrt{2} - 1$

(e) $\dfrac{3}{\sqrt{2}}$

5. The base of a solid is the region bounded by the curves $y = x^3$, $y = 1$, $x = 0$. If the cross sections of the solid perpendicular to the x-axis are squares, then the volume of the solid is equal to

(a) $\dfrac{5}{14}$

(b) $\dfrac{3}{7}$

(c) $\dfrac{9}{14}$

(d) $\dfrac{8}{7}$

(e) $\dfrac{2}{7}$

6. $\int_1^e 16x^3 \ln x\,dx =$

(a) $e - 1$

(b) $2e^3 + 4$

(c) $2e^4 + 4$

(d) $e^4 - 1$

(e) $3e^4 + 1$

7. $\int \tan^4 x\,dx =$

(a) $\tan^5 x + \tan^2 x + C$

(b) $\dfrac{1}{3}\tan^3 x - \tan x + x + C$

(c) $\dfrac{1}{2}\tan x - \dfrac{1}{2}x + C$

(d) $\tan x + \sec^2 x + C$

(e) $2\tan^{-1} x - \tan x + x^2 + C$

8. The arc length of the curve $y = \frac{2}{3}(x^2+1)^{3/2}$ from $\left(0, \frac{2}{3}\right)$ to $\left(1, \frac{4\sqrt{2}}{3}\right)$ is equal to

(a) $\frac{1}{3}$

(b) $\frac{4}{3}$

(c) $\frac{\sqrt{2}}{3}$

(d) $\frac{5}{3}$

(e) $\frac{2}{3}$

9. $\int \cos(5\theta)\cos(3\theta)\,d\theta =$

(a) $\frac{1}{4}\cos(2\theta) + \frac{1}{16}\sin(8\theta) + C$

(b) $\frac{1}{4}\sin(2\theta) - \frac{1}{16}\sin(8\theta) + C$

(c) $-\frac{1}{4}\cos(2\theta) - \frac{1}{16}\cos(8\theta) + C$

(d) $-\frac{1}{4}\sin(2\theta) + \frac{1}{16}\cos(8\theta) + C$

(e) $\frac{1}{4}\sin(2\theta) + \frac{1}{16}\sin(8\theta) + C$

10. The area of the surface generated by revolving the curve $y = \sqrt{4-x^2}$, $-1 \le x \le 1$ about the x-axis is

(a) 4π

(b) 8π

(c) 2π

(d) 5π

(e) 6π

11. If $\alpha = 2\tan^{-1}(-3)$, then $\int_{\alpha}^{0} \frac{1}{\cos x - 3\sin x + 3}\,dx =$ [Hint: Use the substitution $u = \tan\left(\frac{x}{2}\right)$, $-\pi < x < \pi$]

(a) $\ln\left(\frac{9}{5}\right)$

(b) $\ln\left(\frac{7}{5}\right)$

(c) $\ln\left(\frac{8}{5}\right)$

(d) $\ln\left(\frac{6}{5}\right)$

(e) $\ln(2)$

12. If $\frac{1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$, then $A + B + C =$

(a) 2

(b) -1

(c) 0

(d) 1

(e) -3

13. $\int_0^1 \cos^2(\pi x)\,dx =$

(a) $\frac{1}{2}$

(b) 3

(c) 2

(d) 1

(e) $\frac{3}{2}$

14. $\int \frac{\sin x}{\cos x + \cos^2 x}\,dx =$

(a) $\ln|\sin x + \cos x| + C$

(b) $\ln|1 + \sec x| + C$

(c) $\ln|1 + \cos x| + C$

(d) $\ln|1 - \sec x| + C$

(e) $\ln|1 - \cos x| + C$

15. The volume of the solid generated by revolving the region bounded by the curves $y^2 = 2x$, $y = 4$ and $x = 0$ about the x-axis is equal to

(a) 16π

(b) 8π

(c) 128π

(d) 32π

(e) 64π

16. Let S be the solid formed by revolving the region bounded by the curves $y = \sqrt{x}$, $y = 0$, $x = 1$, and $x = 3$ about the x-axis. If c and d are two different numbers in the interval $[1, 3]$ that divide the solid S into three parts of equal volume, then $c^2 + d^2 =$

(a) 10

(b) $\dfrac{10}{3}$

(c) $\dfrac{8}{3}$

(d) $\dfrac{\sqrt{8}}{3}$

(e) $\dfrac{\sqrt{11}}{3}$

17. $\int \frac{8-x}{x^3+4x}\,dx =$

(a) $2\ln|x| + \ln(x^2+4) + \tan^{-1}\left(\dfrac{x}{2}\right) + C$

(b) $\ln|x| - \dfrac{1}{2}\ln(x^2+4) - \tan^{-1}\left(\dfrac{x}{2}\right) + C$

(c) $\ln|x| - \dfrac{1}{2}\ln(x^2+4) + C$

(d) $2\ln|x| - \ln(x^2+4) - \dfrac{1}{2}\tan^{-1}\left(\dfrac{x}{2}\right) + C$

(e) $3\ln|x| + \ln(x^2+4) - \tan^{-1} x + C$

18. $\int_0^8 e^{\sqrt{2x}}\,dx =$

(a) $e^4 - 1$

(b) $4e^4 - 1$

(c) $3e^4 + 1$

(d) $2e^4 + 2$

(e) $2e^4 + 1$

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$x = \sqrt{3}\sin\theta$: $\frac{3}{2}[\theta - \sin\theta\cos\theta]$, back-substituted.
2	**(d)**	$u = \sin t$: $\int(u^{-1/2} - u^{3/2})du = 2\sqrt{\sin t} - \frac{2}{5}\sqrt{\sin^5 t} + C$.
3	**(d)**	Radius $2 - y$, height $(1 - y)(3 + y)$, limits -3 to 1.
4	**(a)**	$x = \tan\theta$: $\int_0^{\pi/4}\cos\theta\, d\theta = \frac{1}{\sqrt{2}}$.
5	**(c)**	$A = (1 - x^3)^2$, $V = \int_0^1(1 - x^3)^2dx = \frac{9}{14}$.
6	**(e)**	By parts, $4x^4\ln x - x^4\Big\vert_1^e = 3e^4 + 1$.
7	**(b)**	$\tan^4 = \tan^2\sec^2 - \sec^2 + 1$, giving $\frac{1}{3}\tan^3 x - \tan x + x + C$.
8	**(d)**	$1 + (y')^2 = (2x^2 + 1)^2$, so $L = \int_0^1(2x^2 + 1)dx = \frac{5}{3}$.
9	**(e)**	$\frac{1}{2}[\cos 2\theta + \cos 8\theta]$ integrates to $\frac{1}{4}\sin 2\theta + \frac{1}{16}\sin 8\theta + C$.
10	**(b)**	$y\sqrt{1 + y'^2} = 2$, so $S = 2\pi\int_{-1}^1 2\, dx = 8\pi$.
11	**(c)**	$u = \tan\frac{x}{2}$: $\int_{-3}^0 \frac{du}{(u-1)(u-2)} = \ln\left\vert\frac{u-2}{u-1}\right\vert\Big\vert_{-3}^0 = \ln\frac{8}{5}$.
12	**(b)**	$1 = A(x + 1)^2 + Bx(x + 1) + Cx$ gives $A = 1, B = -1, C = -1$, sum -1.
13	**(a)**	$\cos^2 = \frac{1+\cos 2u}{2}$: $\frac{1}{2}\left[x + \frac{\sin 2\pi x}{2\pi}\right]_0^1 = \frac{1}{2}$.
14	**(b)**	$u = \cos x$: $-\int\frac{du}{u(u+1)} = \ln\left\vert\frac{u+1}{u}\right\vert = \ln\vert 1 + \sec x\vert + C$.
15	**(e)**	Washers: $\pi\int_0^8(16 - 2x)\, dx = 64\pi$.
16	**(a)**	Total 4π; $c^2 = \frac{11}{3}$, $d^2 = \frac{19}{3}$, so $c^2 + d^2 = 10$.
17	**(d)**	$\frac{2}{x} - \frac{2x}{x^2+4} - \frac{1}{x^2+4}$ integrates to the stated result.
18	**(c)**	$y = \sqrt{2x}$: $\int_0^4 ye^y dy = [ye^y - e^y]_0^4 = 3e^4 + 1$.